The Shell Game: Auditing Allocation in Crosswalked Population Data

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Geographic crosswalking, commonplace in population data workflows, is often described as harmonization of datasets across boundary systems. In practice, crosswalks allocate observations between administrative geographies altering the analytic sample while preserving the variable name. This work introduces two R packages designed to quantify perturbation (ΔX) introduced by crosswalk allocation.The first, shellgame, introduces a conceptual framework and audit protocol for documenting crosswalk assumptions and evaluating how allocation alters the analytic sample. The second, geoDeltaAudit, quantifies perturbation, ΔX(VAR), across allocation steps, making crosswalk-induced sample change measurable.

The demonstration workflow defines geographic membership using Census relationship files and allocates ZIP-level observations to counties using the HUD USPS ZIP–County crosswalk (TOT_RATIO). Development testing additionally evaluates an IPUMS NHGIS block-to-tract interpolation crosswalk derived from block-level geographic relationships. Preliminary results demonstrate measurable perturbation introduced solely through allocation choices prior to statistical analysis. These findings indicate that crosswalk transformations should be treated as allocation procedures and support the use of auditing tools to evaluate crosswalk assumptions in population data workflows.

Allocated outputs imputed via crosswalks are treated as direct measurements. This is the shell game. The variable remains the same but the underlying sample has changed. Methodological artifacts conflate with observations. This structured uncertainty disproportionately impacts boundary communities, yet remains unquantified in standard workflows. shellgame and geoDeltaAudit make these transformations measurable, enabling analysts to document how crosswalk choices shape results used in funding, resource allocation, and policy implementation